

PAPER_1510 — DPM 26-Layer Amplification Factor $A_{26} = \sum_{i=1}^{26} i^6 = 1,307,797,101$ EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket D (Particle Physics) / Bucket A (Foundational) **Date:** June 18, 2026 **Status:** CLOSED — Foundational DPM amplification integer

Observation

PAPER_1155 (DPM 26-Layer Amplification — Particle Masses) derives the universal amplification factor by which $\rho_{SCm} = 7.09 \times 10^{-37} \text{ J/m}^3$ is lifted to particle-mass scales:

$$A_{26} = \sum_{i=1}^{26} i^6 = 1,307,797,101 \quad (\text{exact integer})$$

$$M_{AMU}^{(DPM)} = \rho_{SCm} \times A_{26} \approx 1.627 \times 10^{-27} \text{ kg}$$

Comparison to proton mass $m_p = 1.6726 \times 10^{-27} \text{ kg}$ yields residual -2.04% attributed to the E-crack correction $[SSq] = 0.57$.

UQFF Closed Identity

$$A_{26} = \sum_{i=1}^{D_{crit}} i^6 = 1,307,797,101 \quad \text{EXACT integer}$$

Closed-form polynomial identity:

$$\begin{aligned} A_{26} &= N(N+1)(2N+1)(3N^4 + 6N^3 - 3N + 1) / 42 \quad (\text{Faulhaber, } N=26) \\ &= 26 \cdot 27 \cdot 53 \cdot (3 \cdot 26^4 + 6 \cdot 26^3 - 3 \cdot 26 + 1) / 42 \\ &= 1,307,797,101 \quad \text{EXACT} \end{aligned}$$

Both the direct summation and the Faulhaber polynomial yield the same integer — verified algebraically.

Physical Interpretation

The 26-layer summation amplifies the foundational SCm vacuum density by $\sim 10^9$ to reach the AMU (atomic-mass-unit) scale. Each layer contributes its own i^6 weighting (derived in PAPER_1511) reflecting the structural product of: - SCm density at layer $i \propto i^2$ - UA gradient at layer $i \propto i$ - Background magnetic at layer $i \propto i^3$

Yielding i^6 per layer.

Faulhaber Identity Significance

The closed-form polynomial demonstrates that A_{26} is **not a numerical coincidence** — it is a structural sum of integer powers with a known polynomial identity. The factor of $1/42$ in the denominator (Faulhaber's 6th-power coefficient) places A_{26} in a known sequence of structural sums, lending mathematical rigor.

NOT REPLACEMENT

SM treats the AMU and proton mass as fundamental fit parameters. UQFF supplies a structural product of one foundational density (ρ_{SCm}) and one integer ($A_{26} = 1,307,797,101$) that reproduces the AMU scale within 2%.

Reference

- Source: PAPER_1155 DPM 26-Layer Amplification Particle Masses
- Related: PAPER_1511 (i^6 decomposition), PAPER_062 ($D_{\text{crit}} = 26$), Faulhaber's formula
- Calculator dispatch: `calculate_paradox({"paradox": "dpm_26_layer_amp_a26"})`

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